

Ganesh Chandrasekar

AI/ML Engineer | NLP | LLMs | RAG | Information Retrieval

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PROFESSIONAL SUMMARY

AI/ML Engineer with a Master of Computer Science and experience building retrieval, RAG, and LLM systems. Recent work spans retrieval, reranking, model inference, and evaluation, alongside software engineering experience developing GCP APIs and product workflows.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript/JavaScript, Java

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn, BERT, Embeddings, Sequence Labeling, NER

Retrieval & LLMs: RAG, BM25, Pyserini/Lucene, MedCPT, FAISS, Reranking, LangGraph, MCP, smolagents, vLLM, Ollama, LLM Evaluation

Software & Cloud: REST APIs, Flask, Node.js, Docker, GCP, AWS, Linux, Slurm, Git

PROFESSIONAL EXPERIENCE

Research Assistant

Sep 2023 – Aug 2026

CLaC Lab, Concordia University

Montreal, Canada

- Built retrieval and QA pipelines over **26.8M PubMed documents** using BM25, MedCPT, cross-encoder reranking, evidence selection, and LLM generation for thesis and shared-task research.
- Ran quantized Qwen inference with **vLLM** and Slurm on V100/A100 GPUs, making 32B-model generation practical for large-scale evaluation runs.
- Developed evaluation workflows to separate retrieval, answer quality, grounding, and citation failures, helping compare system variants and identify where errors originated.
- Trained BERT-family, CRF, hierarchical, and PPO models for sequence labeling, and built NER pipelines across biomedical and news text.

Teaching Assistant

Sep 2024 – May 2026

Concordia University

Montreal, Canada

- Led tutorials and office hours for Information Retrieval and Web Search (COMP 479/6791) and Discrete Mathematics (COMP 232), helping students work through assignments and technical questions.

Solution Developer

Jul 2022 – Aug 2023

OrangeScape Technologies Pvt. Ltd.

Chennai, India

- Built **GCP-backed API services** and translated product requirements into workflow rules, validations, and approval paths for Kissflow Procurement Cloud.
- Developed product features across purchasing, approvals, supplier management, and source-to-pay workflows with backend and product teams.

Software Engineering Intern

May 2022 – Jul 2022

Basik Marketing Pvt. Ltd.

Chennai, India

- Developed a real-time esports broadcast overlay with Python and Node.js, automating live match and event-counter updates for the broadcast interface.

PROJECTS

BioEvidence AI: Local-First Biomedical Question Answering

[Live Demo ↗](#)

RAG | Pyserini/Lucene | Local LLMs | Next.js

- Built a full-stack QA application combining Pyserini/Lucene retrieval over a **14.9M-record PubMed index**, adaptive reranking, local LLM generation, and citation validation.
- Integrated PubMed/NCBI, Europe PMC, and ClinicalTrials.gov into a real-time query-to-citation workflow with on-device inference.

Agentic and Non-Agentic Multi-Hop Medical QA

[GitHub ↗](#)

RAG | smolagents | Qwen2.5 | Wikipedia / PubMed

- Built an agentic workflow where Qwen2.5 iteratively searched Wikipedia and decided whether to continue gathering evidence or answer, with a 15-step cap.
- Compared the agentic system with controlled Wikipedia and Wikipedia + PubMed pipelines, examining how tool use and retrieval strategy affected answer quality.

Knowledge-Graph and Hybrid Retrieval for Course Enrollment

[GitHub ↗](#)

RDF/RDFS | SPARQL | Apache Fuseki | LangChain | FAISS

- Built a Concordia course-enrollment knowledge base from CSV/JSON data with **43,291 RDF triples** across **5,258 course URIs**.
- Implemented natural-language question answering with dynamic SPARQL queries and compared graph, FAISS vector, and hybrid retrieval approaches.

Biomedical Negation Detection

[GitHub ↗](#)

PyTorch | BERT | BioBERT | CRF | PPO

- Fine-tuned BERT, BioBERT, and ClinicalBERT for cue/scope BIO tagging, comparing CRF and hierarchical variants designed around class imbalance.
- Trained a PPO-based sequence-labeling policy from annotated spans, improving minority-label detection over the supervised baseline.

EDUCATION

Master of Computer Science (Thesis)

Sep 2023 – Aug 2026

Concordia University

Montreal, Canada

Thesis: *Evidence-Grounded Biomedical Question Answering: Design, Analysis, and Evaluation*

Defended August 2026.

Bachelor of Engineering in Computer Science and Engineering

Jun 2018 – Apr 2022

KCG College of Technology, Anna University

Chennai, India

CGPA: 8.81/10, First Class with Distinction

PUBLICATIONS

TREC 2025 BioGen | *Modular Sparse and Dense Retrieval for Evidence-Constrained Biomedical Question Answering*

BioCreative IX, IJCAI 2025 | *Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering*